

# LARGE VALUES OF WHITTAKER FUNCTIONS FOR $\mathrm{GL}_2$

ABSTRACT. Note written around 2016 where, following Templier [3] and Saha [1], we exhibit large values of  $p$ -adic Whittaker functions for principal series representations of  $\mathrm{GL}_2$ .

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## 1. GENERAL NOTATION

Let  $k$  be a local field. Fix a nontrivial additive character  $\psi$  of  $k$ . When  $k$  is non-archimedean, we assume  $\psi$  is unramified and adopt the accompanying notation  $\mathfrak{o}, \mathfrak{p}, q$  for its maximal order, maximal ideal and residue field cardinality. Let  $\chi_1, \chi_2$  be unitary characters of  $k^\times$  with analytic conductors  $C_1 := C(\chi_1)$ ,  $C_2 := C(\chi_2)$ . Set  $G := \mathrm{GL}_2(k)$ . We use the notation

$$n(x) := \begin{pmatrix} 1 & x \\ & 1 \end{pmatrix}, \quad a(y) := \begin{pmatrix} y & \\ & 1 \end{pmatrix}.$$

Let  $\pi := \mathcal{I}(\chi_1, \chi_2)$  be the induced principal series representation of conductor  $C := C_1 C_2$ .

We write the character group of  $k^\times$  additively. We write  $\alpha$  for the character given by the normalized modulus  $|\cdot|$ . We denote by  $\omega := \chi_1 + \chi_2$  the central character of  $\pi$ . We realize  $\pi$  in its induced model as a space of functions  $f : G \rightarrow \mathbb{C}$  satisfying

$$f(n(x)a(y)zg) = y^{\alpha/2+\chi_1} z^\omega f(g).$$

Here  $z \in k^\times$  identifies with an element of the center of  $G$ , and we use notation such as

$$y^{\alpha/2+\chi_1} := |y|^{1/2} \chi_1(y).$$

We denote in what follows by  $\iota_k$  a constant such as  $\zeta_k(1)$ ,  $\zeta_k(2)$ , or a product of (inverses of) similar quantities; we allow the precise value of  $\iota_k$  to vary from one occurrence to another.

## 2. WHITTAKER FUNCTIONS IN PRINCIPAL SERIES REPRESENTATIONS

**2.1. Schwartz space parametrization.** For  $\Phi \in \mathcal{S}(k^2)$ , define the element  $f_\Phi \in \pi$  by the formula

$$f_\Phi(g) := \int_{z \in k^\times} \Phi(e_2 z g) z^{-\omega} \det(zg)^{\alpha/2+\chi_1}.$$

Thus if

$$g = \begin{pmatrix} * & * \\ c & d \end{pmatrix},$$

then

$$f_\Phi(g) = \det(g)^{\alpha/2+\chi_1} \int_{z \in k^\times} \Phi(cz, dz) z^{\alpha+\chi_1-\chi_2}.$$

This association defines a surjective map  $\Phi \mapsto f_\Phi$  onto  $\pi$ .

**2.2. The Whittaker intertwiner.** Embed  $f \in \pi$  into its  $\psi$ -Whittaker model via the map  $f \mapsto W_f$ , where

$$W_f(g) := \int_{u \in k} f(w_n(u)g) \psi(-u).$$

For  $\Phi \in \mathcal{S}(k^2)$ , set  $W_\Phi := W_{f_\Phi}$ . For  $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$ , we compute that

$$W_\Phi(g) = \det(g)^{\alpha/2+\chi_1} \int_{z \in k^\times} \int_{u \in k} \Phi(z(a+cu), z(b+du)) z^{\alpha+\chi_1-\chi_2} \psi(-u).$$

After the change of variables  $u \mapsto u/z$  followed by  $z \mapsto uz$ , the above becomes

$$W_\Phi(g) = \det(g)^{\alpha/2+\chi_1} \int_{z \in k^\times, u \in k} \Phi(u(az+c), u(bz+d)) (uz)^{\chi_1-\chi_2} \psi(-1/z). \quad (1)$$

**2.3. Specialization to diagonal matrices.** In the special case  $g = a(y) := \begin{pmatrix} y & \\ & 1 \end{pmatrix}$ , the formula (1) specializes to

$$W_\Phi(a(y)) = y^{\alpha/2+\chi_1} \int_{z \in k^\times, u \in k} \Phi(uyz, u) (uz)^{\chi_1-\chi_2} \psi(-1/z),$$

which after the substitutions  $u \mapsto u/yz$ ,  $z = 1/yt$  gives

$$W_\Phi(a(y)) = y^{\alpha/2+\chi_2} \int_{u, t \in k} \Phi(u, ut) u^{\chi_1-\chi_2} \psi(-yt). \quad (2)$$

**2.4. Specialization to matrices in the open Bruhat cell.** The most relevant group element is

$$g = a(y)wn(x) = \begin{pmatrix} & -y \\ 1 & x \end{pmatrix},$$

for which (1) gives

$$W_\Phi(a(y)wn(x)) = y^{\alpha/2+\chi_1} \int_{z \in k^\times} \int_{u \in k} \Phi(u, u(-yz+x))(uz)^{\chi_1-\chi_2} \psi(-1/z).$$

We substitute  $z \mapsto z/y$  to rewrite this as

$$W_\Phi(a(y)wn(x)) = y^{\alpha/2+\chi_2} \int_{z \in k^\times} \int_{u \in k} \Phi(u, u(x-z))(uz)^{\chi_1-\chi_2} \psi(-y/z)$$

and then substitute  $z = x - t$  to obtain

$$W_\Phi(a(y)wn(x)) = y^{\alpha/2+\chi_2} \int_{u, t \in k} \Phi(u, ut) u^{\chi_1-\chi_2} (x-t)^{\chi_1-\chi_2-\alpha} \psi(-y(x-t)^{-1})$$

and finally  $t \mapsto xt$  to conclude that

$$W_\Phi(a(y)wn(x)) = y^{\alpha/2+\chi_2} \int_{u, t \in k} \Phi(u, utx) (ux)^{\chi_1-\chi_2} (1-t)^{\chi_1-\chi_2-\alpha} \psi\left(\frac{-y/x}{1-t}\right). \quad (3)$$

### 3. THE NORMALIZED NEWVECTOR IN A PRINCIPAL SERIES REPRESENTATION

We suppose in this section that  $k$  is non-archimedean.

**3.1. Choice of Schwartz function.** Define  $\Phi \in \mathcal{S}(k^2)$  by the formula

$$\Phi(c, d) := C_2^{1/2} c^{-\chi_1} d^{\chi_2} \cdot \begin{cases} 1_{C_2|c|=1} & C_1 > 1 \\ 1_{C_2|c| \leq 1} & C_1 = 1 \end{cases} \cdot \begin{cases} 1_{|d|=1} & C_2 > 1 \\ 1_{|d| \leq 1} & C_2 = 1. \end{cases}$$

The restriction of  $f_\Phi$  to  $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in K := GL_2(\mathfrak{o})$  is

$$f_\Phi(g) = \det(g)^{\alpha/2+\chi_1} \int_{z \in k^\times} \Phi(cz, dz) z^{\alpha+\chi_1-\chi_2}.$$

This defines a newvector in  $\pi$  with  $L^2$ -norm  $\asymp 1$ . One can see this directly in the induced model (e.g., observe that it is supported on the same  $B \times K_1(C)$ -double coset modulo  $\mathfrak{p}^{n_1+n_2}$  as the one given in Schmidt's note [2], or just verify directly that it satisfies the needed invariance properties and has norm  $\asymp 1$ ). We give another verification below.

We henceforth assume for simplicity that  $C_1, C_2$  are both strictly greater than 1, so that we may simplify the above formula to

$$\Phi(c, d) = C_2^{1/2} c^{-\chi_1} d^{\chi_2} 1_{C_2|c|=1} 1_{|d|=1}.$$

Put another way, we have

$$\Phi(u, ut) u^{\chi_1-\chi_2} = C_2^{1/2} t^{\chi_2} 1_{C_2|u|=1} 1_{|t|=C_2}.$$

**3.2. Sanity check: the Whittaker function at diagonal matrices.** The general formula (2) for the Schwartz function-parametrized Whittaker function at a diagonal element specialized to the choice of  $\Phi$  given above gives

$$W(a(y)) = C_2^{1/2} y^{\alpha/2 + \chi_2} \int_{u, t \in k} t^{\chi_2} 1_{C_2|u|=1} 1_{|t|=C_2} \psi(-yt).$$

The integrand has support of volume  $\asymp 1$ ; the  $u$  integrand is constant, while the  $t$  integrand is a Gauss sum vanishing unless  $|y| = 1$ , in which case it has magnitude  $\iota_k C_2^{-1/2}$ . The change of variables  $t \mapsto t/y$  and  $z := 1/t$  gives

$$W(a(y)) = \iota_k C_2^{1/2} y^{\alpha/2} \int_{z \in k^\times} 1_{C_2|yz|=1} z^{-\chi_2} \psi(-1/z),$$

which by the stated properties of Gauss sums implies that

$$W(a(y)) = \iota_k 1_{\mathfrak{o}^\times}(y),$$

as expected.

**3.3. The Whittaker function at elements of the open Bruhat cell.** As a variant of the formula for  $\Phi(u, ut)$  recorded above, we have

$$\Phi(u, uxt)(ux)^{\chi_1 - \chi_2} = C_2^{1/2} x^{\chi_1} t^{\chi_2} 1_{C_2|u|=1} 1_{|xt|=C_2}.$$

From the general formula for  $W(a(y)wn(x))$  recorded above in (3), it follows that

$$W(a(y)wn(x)) = C_2^{1/2} x^{\chi_1} y^{\alpha/2 + \chi_2} \int_{u, t \in k} t^{\chi_2} (1-t)^{\chi_1 - \chi_2 - \alpha} 1_{C_2|u|=1} 1_{|xt|=C_2} \psi\left(\frac{-y/x}{1-t}\right).$$

The  $u$ -integral is constant with support of volume  $\iota_k C_2^{-1}$ , while the support of the  $t$  integral has volume  $\iota_k C_2/|x|$ , so executing the  $u$  integral while renaming  $t \in k$  to  $z \in k^\times$  gives

$$W(a(y)wn(x)) = \iota_k \left(\frac{C_2|y|}{|x|^2}\right)^{1/2} x^{\chi_1} y^{\chi_2} \int_{z \in k^\times} z^{\chi_2} 1_{|xz|=C_2} (1-z)^{\chi_1 - \chi_2 - \alpha} \psi\left(\frac{-y/x}{1-z}\right)$$

Note that all calculations up until now basically work over an archimedean field, too; we just have to replace conditions like  $1_{|z|=1}$  with a smooth cutoff.

**3.4. Final formula for the magnitude.** In particular, taking absolute values in the final formula from section 3.3 gives

$$W(a(y)wn(x)) \asymp \left(\frac{C_2|y|}{|x|^2}\right)^{1/2} \int_{z \in k^\times} 1_{|z|=C_2/|x|} z^{\chi_2} (1-z)^{\chi_1 - \chi_2 - \alpha} \psi\left(\frac{-y/x}{1-z}\right).$$

It will also be convenient (for later purposes unrelated to the lower bound) to record the result of the substitution  $z \mapsto z^{-1}$ :

$$W(a(y)wn(x)) \asymp \left(\frac{|y|}{C_2}\right)^{1/2} \int_{z \in k^\times} 1_{|z|=|x|/C_2} z^{-\chi_1} (1-z)^{\chi_1 - \chi_2 - \alpha} \psi\left(\frac{zy/x}{1-z}\right).$$

## 4. THE LOWER BOUND

4.1. **Assumptions.** We consider in this section characters whose conductors satisfy

- $C_1 > C_2 > 1$ ,
- $C_1$  is a square,
- $C_2 < C_1^{1/2}$ ,

and  $x, y$  satisfying

- $C_1(C_2/|x|)^2 \leq 1$ ,
- $|y/x|(C_2/|x|)^2 \leq 1$ ,
- $C_2 < |x| < C_1$ .

4.2. **The model case.** It is clear that  $x, y$  satisfying the above assumptions exist; for instance, it suffices to require that

$$|x| = C_2 C_1^{1/2}, |y| = C_2 C_1^{3/2}. \quad (4)$$

4.3. **Calculation.** Under the above assumptions, for any  $z \in k^\times$  with  $|z| = C_2/|x|$  one has  $|1 - z| = 1$  and

$$(1 - z)^{\chi_1 - \chi_2} = \psi(\xi z)$$

for some  $\xi \in k$  satisfying

$$|\xi| = C(\chi_1 - \chi_2) = C_1.$$

Moreover, by a quadratic Taylor expansion  $(1 - z)^{-1} = 1 + z + z^2(1 - z)^{-1}$  (using our assumptions to kill off the contribution of the remainder),

$$\psi\left(-\frac{y/x}{1 - z}\right) = \psi(-y/x)\psi(-yz/x)$$

and so by the first of the above formulas,

$$W(a(y)wn(x)) \asymp \left(\frac{C_2|y|}{|x|^2}\right)^{1/2} \int_{z \in k^\times} 1_{|z|=C_2/|x|} z^{\chi_2} \psi((\xi - y/x)z).$$

The inner integral is a Gauss sum of size

$$\int_{z \in k^\times} 1_{|z|=C_2/|x|} z^{\chi_2} \psi((\xi - y/x)z) \asymp 1_{|\xi - y/x| \cdot C_2/|x| = C_2} C_2^{-1/2} = 1_{|x\xi - y| = |x|^2} C_2^{-1/2}.$$

From our assumption  $|\xi| = C_1 > |x|$  we see that  $|x\xi - y| = |x|^2$  implies  $|y| = C_1|x|$  and thus

$$W(a(y)wn(x)) \asymp \left(\frac{C_1}{|x|}\right)^{1/2} 1_{|x\xi - y| = |x|^2}.$$

Choosing  $x, y$  as in the model case above which satisfy  $|x\xi - y| = |x|^2$  (which is clearly possible), we obtain

$$W(a(y)wn(x)) \asymp \left(\frac{C_1^{1/2}}{C_2}\right)^{1/2}.$$

When is this big? Well, for instance, if  $C_2 = 1$  and  $C_1$  is large. But note that if  $C_1$  and  $C_2$  are the same size (e.g., if the central character is trivial), then the above is actually small, so we don't get values that are obviously large.

## REFERENCES

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